An acid is classed as a substance that produces H^+ ions when it is dissolved in water.

Some acids just split into separate ions for example Hydrochloric (HCl) and Hydrofluoric (HF) acids.

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

Other acids are made from the reaction of a **non-metal oxide** with water when it dissolves. Some of these are carbon dioxide (CO_2), nitrogen dioxide (NO_2) and sulphur trioxide (SO_3).

$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq) \leftrightarrow 2H^+(aq) + SO_4^{-2}(aq)$

The double headed arrow (which should be two arrows on top of each other) shows that the reaction goes both ways.

A base is a substance that produces hydroxide ions (OH^-) when it is dissolved in water.

Metal hydroxides will separate into ions in water.

$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$

Other bases react with water to produce the hydroxide ions

Metal oxides
$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) \rightarrow Ca^{2+}(aq) + 2OH^-(aq)$

Metal carbonates
$CaCO_3(s) + H_2O(l) \leftrightarrow Ca^{2+}(aq) + OH^-(aq) + HCO_3^-(aq)$

Metal hydrogencarbonates
$NaHCO_3(s) + H_2O(l) \leftrightarrow Na^+(aq) + OH^-(aq) + H_2CO_3(aq)$

Ammonia
$NH_3(g) + H_2O(l) \leftrightarrow NH_4^+(aq) + OH^-(aq)$

None of the exercises ask you to show how the hydroxide ions are formed, however it is good to be aware of what happens as you will use it later.

Exercise 37: Acids and Bases

This is just asking you to either name or write the formula of the compound and then classify it as an acid or a base.

If you use the examples you should be ok with this part.

Reactions of Acids and Bases

Acids and bases react in predictable ways. For example, an acid and a hydroxide will always combine to form water and an ionic salt. The ionic salt is formed from the left over reactants.

There are four common types of acid/base reactions.

1. Acid + Hydroxide → Water + Salt

Eg. Hydrochloric acid + Sodium hydroxide → Water + Sodium chloride

$HCl(aq) + NaOH(aq) \rightarrow H_2O(l) + NaCl(aq)$

2. Acid + Oxide → Water + Salt

Eg. Nitric acid + Calcium oxide → Water + Calcium nitrate

$2HNO_3(aq) + CaO(s) \rightarrow H_2O(l) + Ca(NO_3)_2(aq)$

3. Acid + Carbonate → Water + Carbon dioxide + Salt

Eg. Hydrochloric acid + Calcium carbonate → Water + Carbon dioxide + Calcium chloride

$HCl(aq) + CaCO_3(s) \rightarrow H_2O(l) + CO_2(g) + CaCl_2(aq)$

4. Acid + Hydrogencarbonate → Water + Carbon dioxide + Salt

Eg. Nitric acid + Sodium hydrogen carbonate → Water + Carbon dioxide + sodium nitrate

$HNO_3(aq) + NaHCO_3 \rightarrow H_2O(l) + CO_2(g) + NaNO_3$

Another important reaction of acids which is not covered in this section is the reaction of acids with metals

Acid + Metal → Hydrogen gas + Salt
Eg. Hydrochloric acid + Magnesium → Hydrogen gas + Magnesium chloride

$2HCl(aq) + Mg(s) \rightarrow H_2(g) + MgCl_2(aq)$

<u>Exercise 38: Neutralisation</u>
For this exercise, you need to use the general reactions to predict what will be formed in the reaction.

For the first one, you are adding acid to a metal hydroxide, which always produces water and a salt. You need to finish the word equation first, write the symbol equation, and then balance your symbol equation.

Sulphuric acid + Potassium hydroxide → Water + Potassium sulphate

$H_2SO_4(aq) + KOH(aq) \rightarrow H_2O(l) + K_2SO_4(aq)$

Then balance

$H_2SO_4(aq) + \mathbf{2}\ KOH(aq) \rightarrow \mathbf{2}\ H_2O(l) + K_2SO_4(aq)$